Topic 3 – Genetics

Students should:		Maths skills
3.1B	Explain some of the advantages and disadvantages of asexual reproduction, including the lack of need to find a mate, a rapid reproductive cycle, but no variation in the population	
3.2B	Explain some of the advantages and disadvantages of sexual reproduction, including variation in the population, but the requirement to find a mate	
3.3	Explain the role of meiotic cell division, including the production of four daughter cells, each with half the number of chromosomes, and that this results in the formation of genetically different haploid gametes The stages of meiosis are not required	
3.4	Describe DNA as a polymer made up of: a two strands coiled to form a double helix b strands linked by a series of complementary base pairs joined together by weak hydrogen bonds c nucleotides that consist of a sugar and phosphate group with one of the four different bases attached to the sugar	
3.5	Describe the genome as the entire DNA of an organism and a gene as a section of a DNA molecule that codes for a specific protein	
3.6	Explain how DNA can be extracted from fruit	
3.7B	**Explain how the order of bases in a section of DNA decides the order of amino acids in the protein and that these fold to produce specifically shaped proteins such as enzymes**	
3.8B	**Describe the stages of protein synthesis, including transcription and translation:** **a RNA polymerase binds to non-coding DNA located in front of a gene** **b RNA polymerase produces a complementary mRNA strand from the coding DNA of the gene** **c the attachment of the mRNA to the ribosome** **d the coding by triplets of bases (codons) in the mRNA for specific amino acids** **e the transfer of amino acids to the ribosome by tRNA** **f the linking of amino acids to form polypeptides**	
3.9B	**Describe how genetic variants in the non-coding DNA of a gene can affect phenotype by influencing the binding of RNA polymerase and altering the quantity of protein produced**	
3.10B	**Describe how genetic variants in the coding DNA of a gene can affect phenotype by altering the sequence of amino acids and therefore the activity of the protein produced**	

Students should:		Maths skills
3.11B	Describe the work of Gregor Mendel in discovering the basis of genetics and recognise the difficulties of understanding inheritance before the mechanism was discovered	1c 2c, 2e
3.12	Explain why there are differences in the inherited characteristics as a result of alleles	
3.13	Explain the terms: chromosome, gene, allele, dominant, recessive, homozygous, heterozygous, genotype, phenotype, gamete and zygote	
3.14	Explain monohybrid inheritance using genetic diagrams, Punnett squares and family pedigrees	1c 2c, 2e 4a
3.15	Describe how the sex of offspring is determined at fertilisation, using genetic diagrams	1c 2c, 2e 4a
3.16	Calculate and analyse outcomes (using probabilities, ratios and percentages) from monohybrid crosses and pedigree analysis for dominant and recessive traits	1c 2c, 2e 4a
3.17B	Describe the inheritance of the ABO blood groups with reference to codominance and multiple alleles	1c 2c, 2e 4a
3.18B	**Explain how sex-linked genetic disorders are inherited**	1c 2c, 2e 4a
3.19	State that most phenotypic features are the result of multiple genes rather than single gene inheritance	
3.20	Describe the causes of variation that influence phenotype, including: a genetic variation – different characteristics as a result of mutation and sexual reproduction b environmental variation – different characteristics caused by an organism's environment (acquired characteristics)	
3.21	Discuss the outcomes of the Human Genome Project and its potential applications within medicine	
3.22	State that there is usually extensive genetic variation within a population of a species and that these arise through mutations	
3.23	State that most genetic mutations have no effect on the phenotype, some mutations have a small effect on the phenotype and, rarely, a single mutation will significantly affect the phenotype	

Use of mathematics

- Use estimations and explain when they should be used (1d).
- Translate information between numerical and graphical forms (4a).
- Extract and interpret information from graphs, charts and tables (2c and 4a).
- Extract and interpret data from graphs, charts, and tables (2c).
- Understand and use direct proportions and simple ratios in genetic crosses (1c).
- Understand and use the concept of probability in predicting the outcome of genetic crosses (2e).
- Calculate arithmetic means (2b).

Suggested practicals

- Investigate the variations within a species to illustrate continuous variation and discontinuous variation.
- Investigate inheritance using suitable organisms or models.